

EXPERIMENT-6

Aim: Simulating a Local Area Network and LAN topologies.

Instructions for execution:

- 1) We will write a tcl script and simulate it by ns2.
- 2) We begin by specifying the trace files and nam files to be created.
- 3) Define a finish procedure.
- 4) Determine and create the nodes to be used for topology. Here we select 6 nodes: 0,1,2,3,4,5.
- 5) Create links for connecting these nodes.
- 6) Set up the LAN by specifying nodes and assign values for bandwidth, delay, queue type and channel to it.
- 7) Set up the TCP and UDP connection(s) and the FTP/CBR (or any other application) that will run over it.
- 8) Schedule the different events like simulation start and stop, data transmission start and stop.
- 9) Call the finish procedure and mention the time of end of simulation. 10) Execute the script in terminal by command: **ns script_name.tcl**

Program Code:

```
#lan.tcl
#Lan simulation set ns
[new Simulator] #define
color for data flows
$ns color 1 Blue
$ns color 2 Red #open
tracefiles set tracefile1
[open out.tr w] set winfile
[open winfile w]
$ns trace-all $tracefile1
#open nam file set namfile
[open out.nam w] $ns
namtrace-all $namfile
#define the finish procedure
proc finish {} {
    global ns tracefile1 namfile
    $ns flush-trace
    close $tracefile1
    close $namfile
    exec nam out.nam &
    exit 0
}
```

```

#create six nodes set
n0 [$ns node] set n1
[$ns node] set n2
[$ns node] set n3
[$ns node] set n4
[$ns node] set n5
[$ns node] $n1
color Red
$n1 shape box
#create links between the nodes
$ns duplex-link $n0 $n2 2Mb 10ms DropTail
$ns duplex-link $n1 $n2 2Mb 10ms DropTail
$ns simplex-link $n2 $n3 0.3Mb 100ms DropTail $ns
simplex-link $n3 $n2 0.3Mb 100ms DropTail
set lan [$ns newLan "$n3 $n4 $n5" 0.5Mb 40ms LL Queue/DropTail MAC/Csma/Cd Channel]
#Give node position
$ns duplex-link-op $n0 $n2 orient right-down
$ns duplex-link-op $n1 $n2 orient right-up
$ns simplex-link-op $n2 $n3 orient right
$ns simplex-link-op $n3 $n2 orient left
#set queue size of link(n2-n3) to 20
$ns queue-limit $n2 $n3 20 #setup
TCP connection
set tcp [new Agent/TCP/Newreno] $ns
attach-agent $n0 $tcp
set sink [new Agent/TCPSink/DelAck]
$ns attach-agent $n4 $sink
$ns connect $tcp $sink
$tcp set fid_ 1
$tcp set packet_size_ 552
#set ftp over tcp connection
set ftp [new Application/FTP]
$ftp attach-agent $tcp #setup
a UDP connection set udp
[new Agent/UDP] $ns attach-
agent $n1 $udp set null [new
Agent/Null] $ns attach-agent
$n5 $null
$ns connect $udp $null
$udp set fid_ 2
#setup a CBR over UDP connection set
cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$cbr set type_ CBR

```

```
$cbr set packet_size_ 1000
$cbr set rate_ 0.01Mb
$cbr set random_ false
#scheduling the events
$ns at 0.1 "$cbr start"
$ns at 1.0 "$ftp start"
$ns at 124.0 "$ftp stop" $ns at 125.5
"$cbr stop" proc plotWindow
{tcpSource file} {
    global ns set time 0.1 set now
    [$ns now] set cwnd [$tcpSource
set cwnd_] puts $file "$now
$cwnd"
    $ns at [expr $now+$time] "plotWindow $tcpSource $file" }
$ns at 0.1 "plotWindow $tcp $winfile"
$ns at 125.0 "finish"
$ns run
```

Output:

